

TRENT HAINES

1075 Park Place, San Mateo, California 94403

 <https://trentshaines.github.io>

 trentshaines@gmail.com

 <https://www.linkedin.com/in/trentshaines/>

Experience

Roblox

Feb 2024 – Present

Software Engineer II - Backend, Payments Platform & Ledger Infrastructure

San Mateo, California

- Award for most shipped PR's in Economy Org (200+ people) in H1 2025
- Delivered a zero-downtime migration of 10B+ core billing entity records. Led end-to-end data validation, created interface adapters, and refactored legacy systems to maintain seamless compatibility
- Designed and implemented a real-time wallet ledger serving 20K+ transactions per second, backed by a streaming pipeline (Kafka + Flink) materializing ledger roll-ups via Temporal workflows to update hot wallet balances
- Built a drop-in dual-read/dual-write framework adopted as the organization standard across 10 engineering teams. Abstracted away cutover complexity through configurable execution modes, background orchestration, error handling, rollout controls, mismatch telemetry, mismatch configuration, and automatic alerting
- Evolved subscription renewal management from an unreliable table-scan processor to a workflow-driven Temporal orchestrator, improving availability from <99% to >99.99%
- Traced the full service communication network for Payment Search across 3B+ payment records and multiple payment integrations, revealing redundant paths and fragmented data sources. Rearchitected Payment Search to directly communicate with a new Elasticsearch-backed API, eliminating middlemen and establishing a single source of truth
- Redesigned the internal representation of 'product', replacing an inflexible 6-table relational schema with a unified single-table with Protobuf-structured data, serving as the data layer for all real-money transactions on Roblox platform
- Hackweek Project: Built runnable task automation into Roblox Studio (C++/Lua Game Development IDE), equivalent to VSCode's 'tasks.json'. Integrated with the AI assistant to support automatic creation and execution of tasks

Amazon

May 2022 – August 2022

Software Engineering Intern - Fullstack, Route Planning Optimization

Austin, Texas

- Developed an interactive web app enabling Amazon science teams to analyze delivery route execution patterns, guiding data-driven route-planning optimization decisions
- Identified 14 operational metrics (ex: aggregate package weight, delivery density, average stop duration) and implemented end-to-end data pipelines from database queries to front-end visualizations supporting flexible route comparison

Mandiant/FireEye (Part of Google Cloud)

May 2021 – August 2021

Software Engineering Intern - Backend, Infra & Observability

Dallas, Texas

- Modernized container infrastructure for a core threat detection engine through a full ECS → EKS migration
- Created Grafana monitoring & alerting for new EKS engine tracking container performance, throttling, resource use, etc.

University of Texas at Dallas

May 2016 – August 2017

NanoTechnology Research Intern

Richardson, Texas

- Developed applications for thermally actuated nylon artificial muscles, with prototypes incorporated into a successful NASA STTR Phase II grant application
- Synthesized organic aerogels and conducted experiments to test and optimize their material properties

Education

University of Texas at Dallas

August 2019 - May 2023

Master's + Bachelor's of Science in Computer Science

Richardson, Texas

- 4-Year Fast-Track MS/BS
- Full Ride Merit Scholarship
- Computing Scholar's Honors

Relevant Coursework

- Algorithm Analysis
- Distributed Computing
- C++ in UNIX Environment
- Statistical Methods for AI/ML
- Database Systems
- Operating Systems

Projects

Convex Hull Algorithm Visualizer | React, TypeScript

April 2022

- Created a React application that visualizes three different Convex Hull algorithms to help Computational Geometry students understand and compare them through custom data-sets that highlight each algorithm's strengths in various scenarios

Technical Skills

Languages: C#, C++, Python; occasional use of Java/Kotlin, JavaScript/TypeScript

Distributed Systems & Frameworks: ASP.NET Core, gRPC/Protobuf, Temporal Workflows, SQS, Kafka, Flink, Redis

Infrastructure & Observability: Linux, Docker, Prometheus/Grafana, ElasticSearch/ELK